Census structure of floor-attaining local models for the singlet

Supporting computations for arXiv:2608.18886 — deposit note, version 2

Aaron Alai (ORCID 0000-0003-4704-7271), Independent Researcher, Baltimore, MD

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Question

In the moiré-phase-locking mechanism of arXiv:2608.18886 (Sec. 7), each event carries frozen per-station offsets $\left(\xi_A, \xi_B \right)$, a flight time τ and a coupling scale, drawn from a setting-independent density ρ. On 1024 random or stratified offsets at the 161-angle register and tolerance $\varepsilon = 0.01$ the linear program returns $M/M_{\min} = 1.000$, with $M_{\min} = \left(\sqrt{2} - 1 \right)/3$ Hall's floor. The paper states that ρ is optimizer-chosen and that deriving it from a formation process is open. This deposit asks what the floor-attaining ρ looks like, whether it depends on the terrain U, and whether it can be built from the two stations separately.

Method

Four independent floor-attaining solutions were produced with `dither_sweep_rho.py` (NANG = 161, NSH = 32, i.e. 1024 offsets, $\varepsilon = 0.01$, SAVE_RHO = 1): random seeds 0 and 1, stratified seeds 0 and 1. Each has $M/M_{\min} = 1.000$ to twelve digits; the four share no offsets. `census_structure_v2.py` forms the weighted empirical measure over $\left(\xi_A, \xi_B \right)$, pooled over τ and scale, and its circular Fourier plane $\left| F\left(k_a, k_b \right) \right|$ for $|k| \le 6$, with phases; the null is a 400-draw seeded permutation (weights fixed, offsets re-paired). `terrain_factor_v1.py` tests whether the 23-term terrain factors as two screens. `flight_time_perturb_v1.py` / `dither_sweep_perturb_v1.py` re-solve on a perturbed terrain with the same 1024 offsets as random0. `product_census_v3.py` tests, on the random0 banks, whether a census of the form $q\left(\tau, s \right) f\left(\xi_A \right) g\left(\xi_B \right)$ ("product"), or one with ξ_B independent of everything else ("semi"), reproduces the singlet correlators at tolerance ε, against a binned joint census as control; `breed_product_v1.py` with `pipeline_v1.py` breeds a terrain scored by the product-class price and validates the champion at full scale.

Result A — a universal low-order skeleton (four censuses)

census	active atoms	anti-diag $\left(k, -k \right)$ z	diag $\left(k, k \right)$ z	top pixel $\lvert F \rvert$	2nd pixel $\lvert F \rvert$
random0	658	+0.9	+5.8	(2,2) 0.373	(2,0) 0.244
random1	675	−0.7	+6.2	(2,2) 0.352	(2,0) 0.293
strat0	684	−0.7	+5.8	(2,2) 0.349	(2,0) 0.310
strat1	669	−0.9	+7.1	(2,2) 0.384	(2,0) 0.251

Null 95th percentile for the top pixel: 0.17–0.18 on every run. Phases φ of $F\left(k_a, k_b \right)$ (the census contains $|F| \cos\left(2k_a\xi_A + 2k_b\xi_B - \varphi \right)$):

pixel	random0	random1	strat0	strat1	spread	P(4 uniform phases in that arc)
(2,2)	+156.5°	+161.1°	+157.1°	+158.0°	4.5°	8×10^{-6}
(2,0)	+148.2°	+148.9°	+154.5°	+152.3°	6.3°	2×10^{-5}

pixel	random0	random1	strat0	strat1	spread	P(4 uniform phases in that arc)
(4,−2)	−157.9°	−155.1°	−160.1°	−159.1°	4.9°	1×10^{-5}
(2,4)	−84.9°	−82.6°	−68.8°	−83.6°	16°	4×10^{-4}

Statements supported, in descending strength: (1) ρ^* is not a function of $\xi_A - \xi_B$ alone: the anti-diagonal share of the $|k| \le 6$ plane sits at the permutation null on all four censuses. It is not independent of $\xi_A - \xi_B$ either: a per-pixel permutation null for $|F(\,2,\,-2\,)|$ (`pixel_null_v1.py`, same measure, same 400 draws and seed) gives $|F(\,2,\,-2\,)| = 0.127/0.152/0.143/0.124$ against null 95th percentiles $0.117/0.115/0.099/0.106$ ($p = 0.038, 0.005, 0.008, 0.023$), with phase $\psi = +14°/+45°/+50°/+48°$ — a weaker relative-phase term $\cos\,(4\left(\xi_A - \xi_B\right) - \psi)$ at about a third of the sum term, agreeing in phase on three of the four. Whether a relative-phase-only census can attain the floor is addressed by Result F. (2) ρ^* carries a sum term $\cos\,(4\left(\xi_A + \xi_B\right) - \varphi)$, $\varphi \approx 158°$ (so the preferred value of $\xi_A + \xi_B$ is $\varphi/4 \approx 40°$ mod $90°$ in the pinned frame), as its dominant structure and a nonuniform ξ_A marginal $\cos 4\xi_A$; magnitudes and phases agree across two samplers and two seeds sharing no atoms. (3) The A/B asymmetry ((2,0) > (0,2)) is a property of the solve, which pins $a = 0$ and sweeps b, not of the stations. (4) Weight sits at short flights (random0: $\tau = 0.5$ carries 37%, $\tau = 1$ 27%).

Result B — the terrain is not a product of two screens

On `greedy_results.json` (23 terms, $v = 15.2339$): a product of two real cosine screens predicts $w_{mn+} = w_{mn-}$ for every $(\,m, n\,)$; observed ratios run 0.05 to 183. A rank-1 complex-screen fit within each sign class misses by 2.3–2.8× rms. Difference beats ($k = m - n$) carry 72% of the weight. This is a statement about the greedy-found terrain only.

Result C — terrain perturbation (matched control: random0, same 1024 offsets)

run	terrain change	$M - M_{\min}$	(2,2) $\lvert F\rvert$, φ	(2,0)	(4,−2)	new in top 5
random0 (control)	none	2×10^{-14}	0.373, +156.5°	0.244, +148.2°	0.184, −157.9°	—
A	(2,2,+) phase +45°	5×10^{-8}	0.422, +118.6°	0.258, +127.6°	0.199, −169.7°	(2,−2) 0.237, (0,4) 0.189
B	(2,2,+) removed	3×10^{-14}	0.307, −144.5°	—	0.206, −165.6°	(2,4) 0.285, (0,2) 0.253

The floor does not depend on the (2,2,+) term; the census does. Under a +45° shift of the term's phase the (2,2) pixel moves by −38° (the sign the convention $\sin\,(\,2k\lambda - \phi\,)$ dictates). With the term removed a (2,2) pixel persists at 0.31 at a different phase, from harmonic mixing of the remaining terms, so the pixel tracks the term but is not solely owned by it. The census skeleton is a function of the terrain; the price is not.

Result D — station-local censuses do not reach the floor

On the random0 banks (`product_census_v3.py`, NB = 32 bins, 14 starts per class — uniform, ρ^*-marginals, random0–11 — $\varepsilon = 0.05$ and 0.01; transcript `product_census_v3_random0_NB32_oct_17_16.txt`): no product census and no semi census reproduces the correlators at $\varepsilon = 0.01$ from any start (the script constrains $\left|E - E_{\mathrm{QM}}\right| \le \varepsilon + u$; the final slacks u are 2.2–4.9% (semi) and 3.1–12.4% (product), i.e. the correlators miss their targets by 3–13%), while the binned joint census does (1.119× the floor, the excess being the binning). At $\varepsilon = 0.05$ the station-local classes are feasible and pay 2.20 (semi) and 3.07 (product) times the floor; the binned joint control hit

its time limit there. An earlier NB = 16 run (`product_census_v2.py`, 6 starts; joint 1.59×, station-local classes infeasible at $\varepsilon \leq 0.02$) is recorded in the author's notes; its output was not retained and it is not part of this deposit. Fourier checks on the best product censuses confirm they are honest products ($|F(2,2)| \approx |F(2,0)|\,|F(0,2)|$).

Result E — a terrain bred to favour station-local censuses does not change this

`breed_product_v1.py` from the greedy terrain (breeding scale NSH = 8, NANG = 9, $\varepsilon = 0.10$) stopped after five rounds with no improving move: joint 1.631×, product 1.927× the floor, ratio 1.18 — "promising" under the pre-registered rule (feasible and ratio ≤ 1.3). Full-scale validation of the champion (NANG = 161, 1024 random offsets, NB = 32, 12 starts): at $\varepsilon = 0.05$ joint 1.261, semi 2.492, product 2.864; at $\varepsilon = 0.01$ joint 1.591, semi and product infeasible. The breed improved the product class only relative to the joint class at coarse scale, by degrading the joint; it never brought a product census near the floor and the gain vanished at full resolution. Stage B (minimal start) was not run; the exclusion reads "from the greedy start".

Result F — the census does not factorize in sum and difference coordinates either

`product_census_v4.py` (v3 + COORDS = sumdiff: bins in $S = \xi_A + \xi_B$ and $D = \xi_A - \xi_B$), random0 banks (self-check 12 digits), NB = 32, 14 starts per class, seed 7; author's machine, 17,655 s; transcript `product_census_v4_sumdiff_random0_10_21_26.txt`.

ε	**joint (32×32 bins)**	**semi (D independent of all)**	**product $f(S)\,g(D)$**
0.05	time limit	1.965 (14/14 feasible)	2.749 (12/14 feasible)
0.01	1.107	none feasible from 14 starts; targets missed by 1.7–3.0%	none feasible from 14 starts; targets missed by 2.2–3.9%

Against the station-local classes on the same banks (Result D: 2.20 / 3.07 at $\varepsilon = 0.05$; misses of 3–13% at $\varepsilon = 0.01$), the sum/difference split is about 10% cheaper where feasible and misses by roughly three times less where not: the census is closer to separable in (S, D) than in $\left(\xi_A, \xi_B\right)$, but not separable. A census depending on D alone is the sub-case \$f(S) = \$ const of the product class, so it is covered by the same statement. In Fourier terms, the dominant $(2,2)_{AB}$ pixel is the pure-S harmonic $(2,0)_{SD}$; the other four skeleton pixels — $(2,0)_{AB} \to (1,1)_{SD}$, $(4,-2)_{AB} \to (1,3)_{SD}$, $(2,4)_{AB} \to (3,-1)_{SD}$, $(4,2)_{AB} \to (3,1)_{SD}$ — are cross terms at 0.18–0.31, which no product in (S, D) can produce. Relative to the pre-registered decision rule this outcome lies between "not separable" and "separable": infeasible at $\varepsilon = 0.01$, but markedly nearer than the station-local class. The best feasible product census at $\varepsilon = 0.05$ drives $|F(2,2)|$ to 0.82 (S nearly fixed).

Standing conclusion

On the paper's terrain, on four independent censuses, and on a terrain bred to favour station-local censuses, no product census reproduces the singlet at $\varepsilon = 0.01$ from any start, and where feasible it pays 2–3× the floor. The frozen offsets of the two stations must be correlated at the source, and the correlation is not separable in the sum and difference of the offsets either (Result F): the sum carries the dominant structure, with the difference conditioned on it. This does not touch the disc–lattice interaction as a reading of U or of D; it excludes it as the origin of ρ. Nothing here derives ρ.

Caveats

The exclusion is computational and "on these terrains", not a theorem; the theorem-level statement is Bell's (station-side processing is $A(a, \lambda)$). The (2,2) pixel tracks the (2,2,+) term but is not solely owned by it. Infeasibility

statements for the station-local and sum/difference classes are “from every start” of an alternating minimization (14 starts), not certificates. Results A (random0), B, the per-pixel null (all four censuses) and the perturbation module’s identity check were reproduced on a second machine; the remaining runs are from the author’s workstation with the scripts’ internal self-checks (stored-ρ price reproduced to twelve digits before each run). `product_census_v1`/`v2` are withdrawn (v1 had a normalization error) and are not part of this deposit.

Not included in this deposit

The minimal-start terrain breed; the Hamiltonian residence-law, profile-spectrum, jitter-damping and asymmetric-frame questions. These will appear in a later version of this record if run. The explicit Fourier series of the census in (S, D), the conditional structure of D given S, and higher-rank separable classes are in progress and will be deposited separately.

Files

`census_structure_v1.py`, `census_structure_v2.py`, `terrain_factor_v1.py`, `flight_time_perturb_v1.py`, `dither_sweep_perturb_v1.py`, `product_census_v3.py`, `product_census_v4.py`, `breed_product_v1.py`, `pipeline_v1.py`, `RUN_PIPELINE.bat`, `check_folders_v1.py`; `greedy_results.json`; `rho_nsh32_{random0,random1,strat0,strat1}_eps0.01_nang161.json`; `rho_perturb_2_2_1_{shift_45,weight_0}_nsh32_random0_eps0.01_nang161.json`; `banks_nsh32_random0_nang161_nw2048.npz`; `product_census_v3_random0_NB32_oct_17_16.txt`, `product_census_v3_rho_nsh32_random0_eps0.01_nang161.json`, `product_census_v4_sumdiff_random0_10_21_26.txt`, `pixel_null_v1.py`, `pixel_null_v1_run_2026-10-21.txt`; `breedrun/A_breed_greedy/{breed_product_results.json, bred_product_champion.json}`; `breedrun/VA_validate_greedy/`; `pipeline_FINAL.txt`; run transcripts. Every number quoted above is produced by the named script from the named input; the scripts assert the stored price before reporting.